

1 nf-core/funcscan: A Nextflow pipeline to identify the 2 biosynthetic potential and resistome of bacterial 3 (meta)genomes

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Submitted: 01 January 1970

Published: unpublished

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Summary

20 Genome-mining of bacterial DNA enables the discovery of antimicrobial resistance-related
21 genes, genes required for the biosynthesis of low molecular weight natural products, and
22 other specialised metabolites. However, execution of the multiple bioinformatic tools used in
23 screening analyses remains inefficient due to heterogenous software interfaces, reporting, and
24 formatting of the output files of similar tools, which limits scalability of such analyses.

25 nf-core/funcscan is a portable and reproducible open source Nextflow bioinformatics pipeline for
26 the screening of microbial functional features from assembled contigs or genomes. The pipeline
27 executes up to 13 tools to simultaneously identify antimicrobial peptides, antibiotic resistance
28 genes, biosynthetic gene clusters, carbohydrate-activate enzymes, and perform taxonomic
29 classification of partial or full genomes. To facilitate efficient results comparison and evaluation,
it supports cross-tool output file standardisation and aggregation.

Statement of need

32 Researchers often use multiple tools to ensure maximum detection sensitivity during genomic
33 screening for potential gene candidates, as each tool uses different search algorithms and
34 microbial metabolite databases. However, heterogenous installation, inputs, and execution
35 interfaces of these stand-alone tools impedes scalability, and decreases reproducibility due to
36 user-error when executed manually. Additionally, each tool often has its own unique output
37 formats, making cross-comparison of results between tools and databases non-trivial, and again
38 requiring inefficient manual postprocessing and inspection.

39 This necessity for manual execution and postprocessing of heterogenous outputs impacts the
40 discovery of new drugs. For example, antibiotics are typically derived from naturally evolved,

bacterially-produced, low molecular weight natural products, and the rate of discovery of novel molecules has seen recent plateauing. In combination with an explosion in the evolution of multidrug resistant bacteria (Perry et al., 2016; Rascovan et al., 2016; Ventola, 2015), and a lack of global surveillance both in healthcare and agriculture, this is contributing to a major threat to global health (*Global Antimicrobial Resistance and Use Surveillance System (GLASS) Report 2022*, 2022; Murray et al., 2022). Therefore high-throughput and scalable approaches are needed to allow the rapid identification of metabolites from novel sources, as well as live monitoring of the spread of antibiotic resistance within microbial populations.

Here, we present nf-core/funcscan, a Nextflow (Di Tommaso et al., 2017) pipeline following nf-core best practices (Ewels et al., 2020; Langer et al., 2025) for the automated and in-parallel screening of different functional gene groups with multiple tools and databases. The pipeline currently supports detection of antimicrobial peptide (AMPs) genes, antimicrobial resistance genes (ARGs), biosynthetic gene clusters (BGCs), and carbohydrate-active enzyme gene clusters (CGCs).

State of the field

Previous efforts to scale up the predictive power of different tools for functional gene prediction include metannotator (Gurbich et al., 2025), bacannot (Almeida et al., 2023), SqueezeMeta (Tamames & Puente-Sánchez, 2019), MetaErg (Dong & Strous, 2019), METABOLIC (Zhou et al., 2022), ARGs-OAP (Yin et al., 2022), and PathoFact (De Nies et al., 2021) (Table 1). However, to our knowledge, these typically focused on singular gene categories or groups (e.g. antimicrobial resistance), aim to be ‘end-to-end’ pipelines including read preprocessing and assembly, or do not provide important contextual information about the potential hits (such as taxonomic information).

Table 1: Comparison of nf-core/funcscan with other related pipelines for ARG, AMP, and BGC discovery. Parentheses indicate either unspecific gene screening or partly fulfilled criteria.

Feature	funcscan	metannotator	bacannot	PathoFact	SqueezeMeta	MetaErg	ARGs-OAP
ARG screening	+	+	+	+	(+)	(+)	+
AMP screening	+	—	—	—	(+)	(+)	—
BGC screening	+	+	—	—	(—)	(—)	—
CAZyme screening	+	+	—	—	—	—	—
Taxonomic assignment of contigs	+	—	—	(—)	+	+	—
Results summary	+	+	+	(+)	+	+	—
Container support (Docker, Singularity)	+	+	+	—	—	+	(—)
Modularity	+	+	+	+	(+)	—	—
One-click installation	+	+	+	—	(—)	—	—
Local installation possible	+	+	+	+	+	+	—
Web-based execution possible	(+)	(+)	(+)	—	—	—	—
Software reviewing	+	+	—	—	—	—	—
Automated unit tests	+	+	(—)	(—)	—	—	—
License	MIT	Apache-2.0	GPL-3.0	GPL-3.0	GPL-3.0	AFL	AFL

Extensive command-line knowledge and manual installation of software dependencies are also often required to run many of these existing pipelines. This can preclude use by biochemists, biologists, etc. who typically have limited computational training. In contrast, nf-core/funcscan

67 aims to reduce complexity by screening from already assembled sequences, and end on
68 aggregation of the screening results, through multiple methods for execution.

69 The main factors that distinguish nf-core/funcscan from the most similar pipeline, metannotator,
70 are: support for metagenomic assembly input (rather than just genomes); automated taxonomic
71 classification of contigs; more ARG tools; standardised prediction output; and confirmed
72 executable on other infrastructure than HPCs.

73 Workflow overview

74 nf-core/funcscan simultaneously predicts AMPs, ARGs, BGCs as well as CGCs from input
75 partial or full (meta)genomic sequences. Output files from the tools of each of the categories
76 are aggregated and standardised for easy cross-comparison (Fig. 1).

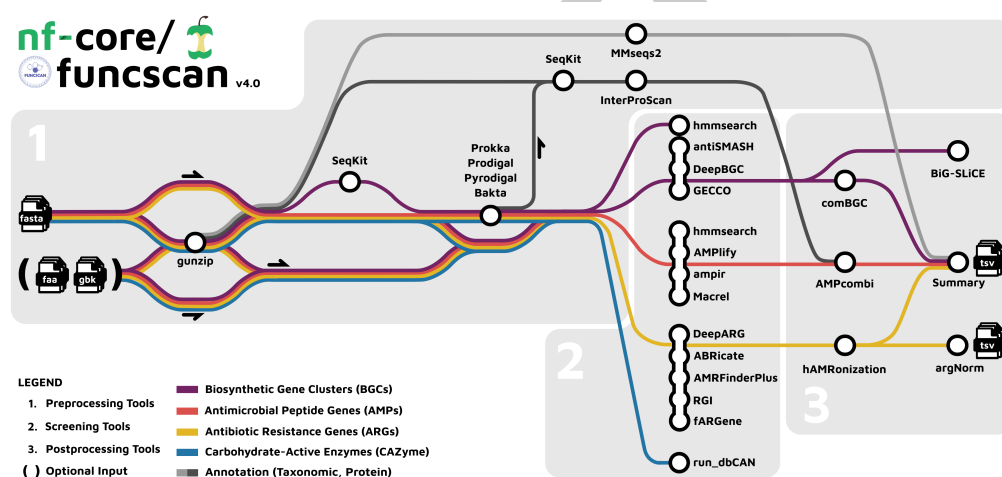


Figure 1: Workflow overview of nf-core/funcscan. (1), genomic sequences are prepared and annotated with one of four open reading frame annotation tools. Two additional classification workflows can be used to classify contigs taxonomically (light gray) or obtain additional protein domain information (dark gray). (2), depending on user-choice, the biosynthetic gene cluster (BGC, purple), antimicrobial peptide genes (AMP, orange), antibiotic resistance genes (ARG, yellow), or carbohydrate-active enzymes (CAZyme, blue) workflows with their customisable parameters are executed. (3), the results of all tools for each gene category are aggregated and saved in a human- and machine-readable tabular format.

77 Input preprocessing and open reading frame annotation

78 The pipeline takes a two- to five-column tabular sample-sheet as input (comma-separated,
79 CSV format). This sample-sheet contains sample names, paths to (meta)genomic FASTA files
80 and optionally pre-generated amino-acid FASTA, GFF, or GBK format annotation files.

81 Preprocessing steps reduce runtime by removing too-short sequences with Seqkit (Shen et al.,
82 2024), when they may produce no biologically meaningful results. Open reading frames are
83 optionally predicted from the preprocessed sequences by one of four prokaryotic annotation
84 tools: Bakta (Schwengers et al., 2021), Prodigal (Hyatt et al., 2010), Prokka (Seemann, 2014),
85 and Pyrodigal (Larralde, 2022).

86 When required, the pipeline will download required screening-tool databases automatically for
87 the user, and made available for future pipeline runs to minimise runtime and network traffic.

Gene screening and taxonomic classification

Users choose to scan genomic sequences in parallel with up-to four dedicated subworkflows for AMPs, ARGs, BGCs, and CGCs. User can apply currently up to a total of 13 gene identification tools:

- ARGs: ABRicate (Torsten Seemann, 2020), AMRFinderPlus (Feldgarden et al., 2019, 2021), DeepARG (Arango-Argoty et al., 2018), fARGene (Berglund et al., 2019), RGI (Alcock et al., 2023)
- BGCs: antiSMASH (Blin et al., 2025), DeepBGC (Hannigan et al., 2019), GECCO (Carroll et al., 2021), hmmsearch (Eddy, 2011)
- AMPs: ampir (Fingerhut et al., 2021), AMPLify Li et al. (2022), hmmsearch, Macrel (Santos-Júnior et al., 2020)
- CAZymes: run_dbCAN (Zheng et al., 2023)

To provide users information about potentially suitable hosts for downstream experiments, e.g. heterologous expression systems (Porse et al., 2018), an additional optional parallel workflow can taxonomically classify input contigs with MMSeqs2 (Mirdita et al., 2021). Optionally, generic protein domains and families can be further annotated with InterProScan (Jones et al., 2014).

Pipeline parameters can be adjusted by userwritten- or nf-core GUI (<https://nf-co.re/launch>)-generated Nextflow parameter files, or command-line arguments.

Aggregation of screening results

nf-core/funcscan integrates dedicated tools to aggregate and standardise heterogenous output formats (and gene labels) of multiple screening tools into a single human- and machine-readable tables in CSV format per gene type. nf-core uses hAMRonization (Mendes et al., 2024) for ARGs, AMPcombi (Herbst et al., 2025) for AMPs, and a custom script 'comBGC' for BGC tool output. These summaries are finally optionally complemented with results from the taxonomic classification workflow.

Reproducibility and scalability

All nf-core pipelines utilise software environments (from the Bioconda project, Grüning et al., 2018) or containers (e.g. Docker, Singularity, primarily from the Biocontainers project, Veiga Leprevost et al., 2017) for each integrated tool. This provides the advantage of isolating the dependencies of all workflows from each other reducing installation problems. The pipeline is thus easy to install with few minimum dependencies - Nextflow itself, and one of Nextflow-supported container/software environment management systems. For further portability, nf-core provides integrated configurations for more than 150 institutional computational infrastructure (e.g. HPCs) via nf-core/configs (<https://nf-co.re/configs>). Users on these infrastructure thus can run the pipelines with no-set up via a single parameter.

Research impact statement

nf-core/funcscan has an active user community of scientific users and developers on the nf-core Slack and GitHub (<https://nf-co.re/join>). For example, the pipeline received the contribution of the CAZyme screening from community members outside of the original developers. User discussions and support on the pipeline and on related research topics occur on the nf-core Slack workspace. This illustrates the public interest and proactive efforts from scientific users to use, maintain, and improve the pipeline. The pipeline is also already actively being used in research Janák et al. (2026).

AI usage disclosure

No generative AI tools were used in the development of this software, the writing of this manuscript, or the preparation of supporting materials.

Acknowledgements

We thank Vedanth Ramji for adding argNorm to the ARG subworkflow. A full list of nf-core community members is available at <https://nf-co.re/contributors/>. We thank Martin Klapper and Rosa Herbst for helpful feedback on relevant BGC and AMP properties during comBGC and AMPcombi development. J.F. received a fellowship from the International Leibniz Research School (under the head of the Jena School for Microbial Communication, JSMC).

This project was funded by grants from the Werner Siemens Foundation (Paleobiotechnology to C.W. and P.S.) and the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation, under Germany's Excellence Strategy – EXC 2051 – Project-ID 390713860 to C.W. and P.S.). J.A.F.Y. and C.W. were funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – project number 460129525 (NFDI4Microbiota, FlexFund project EnterArchaeo). J.A.F.Y. and C.W. were supported by the Max Planck Society. This work was supported by the BMBF-funded de.NBI Cloud within the German Network for Bioinformatics Infrastructure (de.NBI) (031A532B, 031A533A, 031A533B, 031A534A, 031A535A, 031A537A, 031A537B, 031A537C, 031A537D, 031A538A).

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